

Appendix Note: Dual-Role Figure and Firewall Mechanism

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1 Scope and inputs

This note documents the reconstruction of the dual-role Figure 2 and the firewall table without running, sourcing, or modifying a simulation. The generation script reads only these existing result artifacts:

The central DGP has Z_{t-1} as a confounder of D_t and Y_t , while contemporaneous Z_t is a collider of D_t and Y_t . The estimand is the contemporaneous treatment effect $\beta = 1$. The eight-model result artifact uses $N = 100$, $T = 30$, a burn-in of 100 periods, and 500 replications in each of its 10 scenarios. Its displayed models are pooled short/long, TWFE short/long, ADL short/long without fixed effects, and ADL short/long with unit and time fixed effects. The Figure 2 subset fixes $\sigma_{\alpha_Z} = 0.5$ and displays TWFE short, TWFE plus Z_{t-1} , ADL+FE, and ADL+FE plus Z_{t-1} .

The firewall artifact varies (ρ_Z, ρ_Y) over 12 scenarios and reports 500 replications per scenario. It compares the long TWFE specification with the long ADL+FE specification. The existing dual-role report identifies this as the same $N = 100$, $T = 30$ family; the Task 10 inventory records 500 replications but leaves the firewall row's N and T as source metadata. No claim here uses a new draw or an unrecorded simulation run.

2 Monte Carlo construction and fidelity check

Bias is the simulated mean of $\hat{\beta} - 1$; RMSE is $\sqrt{\text{mean}[(\hat{\beta} - 1)^2]}$. For the eight-model raw file, the Monte Carlo standard error (MCSE) is the sample standard deviation of $\hat{\beta}$ across replications divided by $\sqrt{500}$. The plotted 95% Monte Carlo uncertainty interval is mean bias plus or minus $1.96 \times \text{MCSE}$. These quantify simulation noise in the estimated mean bias; they are not confidence intervals for an estimator in a single dataset. The firewall CSV supplies its published MCSEs, so the table applies the same formula to those reported quantities.

The raw recalculation reproduces the published eight-model bias means and MCSEs to numerical precision before any presentational change. The maximum discrepancy is recorded in the validation artifact rather than silently replaced.

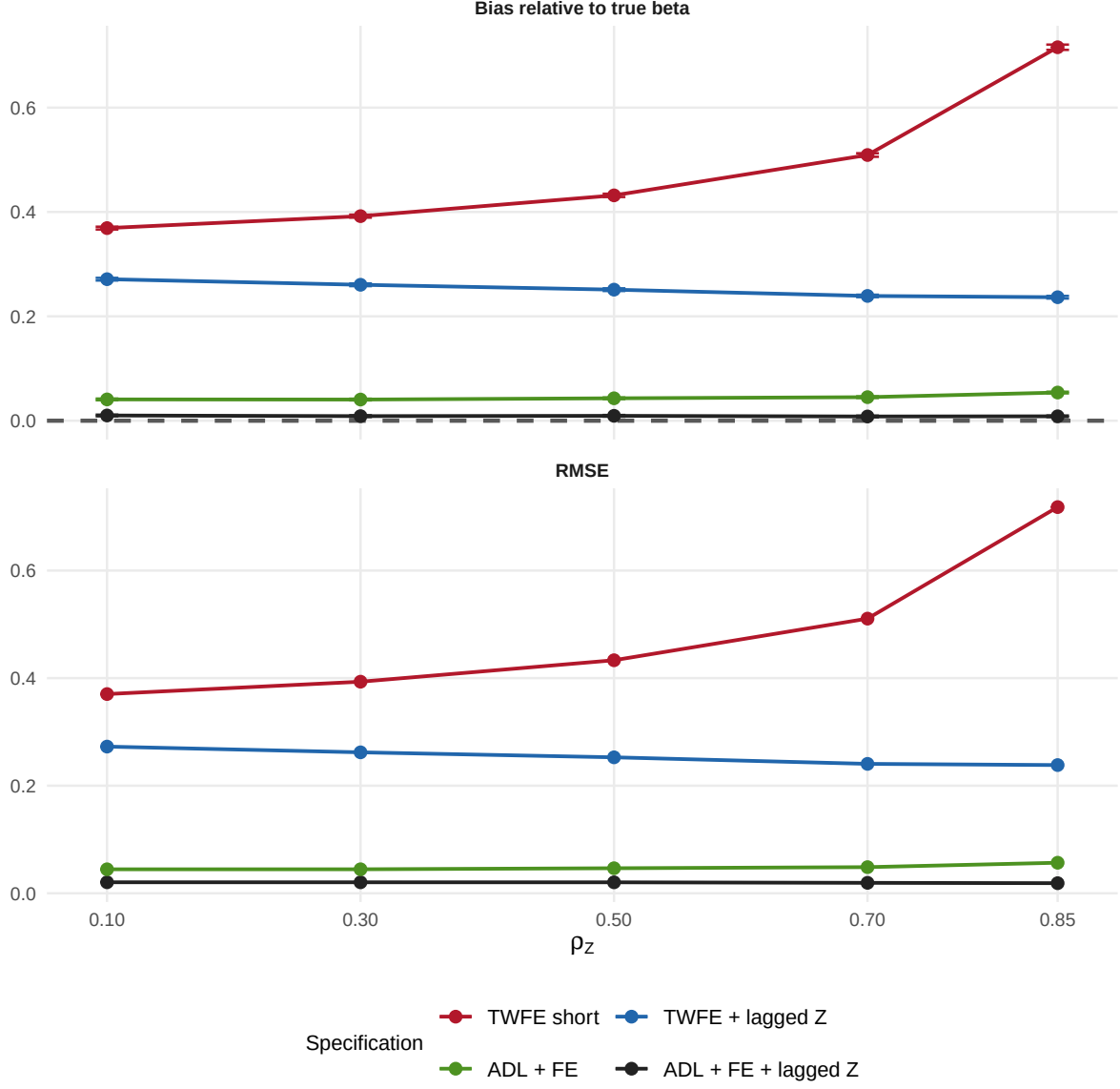
Table 1: Source result artifacts used to generate the Task 11 figure and table.

source_artifact	md5
simulations/dual_role_z/results/sim_dual_role_z_8models_raw.csv	25cca514b6ec2c33b409f0c94b648742
simulations/dual_role_z/results/sim_dual_role_z_8models_results.csv	eb5ac53a9dd902b35764d40de1b28629
simulations/dual_role_z/results/sim_dual_role_z_firewall_results.csv	3a5a2bc3ac6d9abeff4197f9ceb19cbb

Table 2: Reconstruction checks. The figure/table generator stops if recalculated means or MCSEs fail the stated tolerance.

check	value	threshold_or_expected	pass
raw replication count per model-scenario	500	500	TRUE
published replication count per scenario	500	500	TRUE
recalculated bias means reproduce published summary	0	$\leq 1\text{e-}10$	TRUE
recalculated MCSEs reproduce published summary	0	$\leq 1\text{e-}10$	TRUE
firewall grid has 12 scenarios	12	12	TRUE

3 Reconstructed Figure 2



Error bars in the bias panel are 95% Monte Carlo uncertainty intervals for the simulated mean bias (not estimator confidence interval)

Figure 1: Dual-role DGP: bias relative to the true contemporaneous treatment effect and RMSE across collider persistence. The DGP has Z_{t-1} as a confounder and Z_t as a contemporaneous collider; the estimand is $\beta = 1$. The displayed subset sets $\sigma_{\alpha_Z} = 0.5$, with $N = 100$, $T = 30$, a burn-in of 100 periods, and 500 Monte Carlo replications per scenario. Models are TWFE short, TWFE plus Z_{t-1} , ADL+FE, and ADL+FE plus Z_{t-1} ; FE models include unit and time effects. Error bars are 95% Monte Carlo uncertainty intervals for simulated mean bias, not estimator confidence intervals.

4 Firewall mechanism table

Table 3: Firewall mechanism in the dual-role DGP (true contemporaneous treatment effect $\beta = 1$). Each cell reports simulated mean bias and its 95% Monte Carlo uncertainty interval in brackets; these are uncertainty intervals for the simulated mean, not confidence intervals for an estimator. The absolute-bias reduction is $|\text{Bias}_{\text{TWFE}+Z_{t-1}}| - |\text{Bias}_{\text{ADL}+\text{FE}+Z_{t-1}, Y_{t-1}}|$. The existing firewall result artifact reports 500 replications per scenario; the associated dual-role design has $N = 100$, $T = 30$, and a burn-in of 100 periods. Models include unit and time fixed effects.

ρ_Y	ρ_Z	TWFE + Z_{t-1} : bias [MC 95% interval]		ADL + FE + Z_{t-1} : bias [MC 95% interval]		Absolute-bias reduction
0.0	0.30	0.0002	[-0.0013, 0.0017]	0.0076	[0.0060, 0.0092]	-0.0074
0.0	0.50	0.0006	[-0.0010, 0.0021]	0.0081	[0.0065, 0.0098]	-0.0076
0.0	0.70	-0.0010	[-0.0025, 0.0004]	0.0065	[0.0049, 0.0081]	-0.0055
0.0	0.85	-0.0008	[-0.0023, 0.0007]	0.0065	[0.0049, 0.0081]	-0.0057
0.5	0.30	0.2617	[0.2592, 0.2643]	0.0095	[0.0079, 0.0112]	0.2522
0.5	0.50	0.2511	[0.2486, 0.2536]	0.0093	[0.0078, 0.0109]	0.2418
0.5	0.70	0.2384	[0.2361, 0.2408]	0.0079	[0.0064, 0.0094]	0.2305
0.5	0.85	0.2363	[0.2338, 0.2388]	0.0088	[0.0072, 0.0104]	0.2275
0.8	0.30	0.2924	[0.2875, 0.2974]	0.0051	[0.0035, 0.0066]	0.2874
0.8	0.50	0.2615	[0.2566, 0.2664]	0.0042	[0.0027, 0.0057]	0.2573
0.8	0.70	0.3485	[0.3432, 0.3538]	0.0038	[0.0024, 0.0053]	0.3447
0.8	0.85	0.3996	[0.3948, 0.4045]	0.0076	[0.0060, 0.0091]	0.3920

5 Interpretation and limitations

For this DGP, conditioning on the lagged outcome and lagged control is associated with a markedly smaller simulated bias than the long TWFE specification when outcome persistence is positive. When $\rho_Y = 0$, the long TWFE bias is itself approximately zero, so the contrast is not evidence that the lagged outcome universally improves a specification. It is consistent with the displayed mechanism in which outcome persistence transmits the inherited path; it does not apportion the contrast between d-separation, dynamic conditional-mean specification, or finite- T estimator behavior.

The evidence is conditional on this linear, homogeneous DGP and its stated grid. The firewall result is a published summary CSV rather than a raw-replication artifact, so its MC intervals are reconstructed from reported MCSEs rather than recalculated from individual draws. Neither the intervals nor the figure establish estimator-level sampling coverage.